



The effect of executive function on health related quality of life in children with self-limited epilepsy with centrottemporal spikes

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ARTICLE INFO

Keywords:

Self-limited epilepsy with centrottemporal spikes SeLECTS
Health-related quality of life HRQoL
Executive functions EFs

ABSTRACT

Aim: The current study aims to investigate the effect of Executive Functions (EFs) on Health Related Quality of Life (HRQoL) in a cohort of children with self-limited epilepsy with centrottemporal spikes (SeLECTS) and to identify possible factors that impact HRQoL specifically related to epilepsy-related variables and EFs skills.

Material and method: The Pediatric Quality of Life Inventory 4.0 Generic Core Scales (PedsQL) and The Behavior Rating Inventory of Executive Function (BRIEF-2 and BRIEF-P) were completed by the parents of 129 patients with SeLECTS. Demographic variables and epilepsy-related variables were collected.

Results: Our sample performed in the average range across all the subscales and summary scores of the PedsQL and performed in the normal range of the BRIEF questionnaire. We observed that a lower functioning in EFs was associated with lower overall HRQoL scores. We explored the relationship between epilepsy characteristics and scores on the PedsQL. We found that the use of antiseizure medications (ASMs), longer duration of the treatment, and a higher seizure frequency were associated with a lower HRQoL. Moreover, we observed that executive dysfunction was a significant predictor of reduced HRQoL.

Conclusion: Our results suggest the importance of the identification of patients with SeLECTS with a high level of risk for a poor HRQoL. We may now add executive dysfunction to the list of known risk factors for poor HRQoL in children with SeLECTS, along with such factors as seizure frequency, recent seizures, use of ASMs and longer duration of therapy. The early identification of children with SeLECTS at risk of a poor HRQoL could allow the activation of adequate interventions.

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<https://doi.org/10.1016/j.yebeh.2023.109607>

Received 4 September 2023; Received in revised form 11 November 2023; Accepted 21 December 2023

Available online 25 January 2024

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1. Introduction

In accordance with the International League Against Epilepsy (ILAE) definition [1], epilepsy is “a disorder of the brain characterized by an enduring predisposition to generate epileptic seizures and by the neurobiological, cognitive, psychological, and social consequences of this condition”. This definition underlines that epilepsy affects not only the physical functioning but also the neurocognitive, emotional, and social functioning of an individual. In fact, in recent years, objectives of treatment in patients with epilepsy are not only focused on seizures control but also on psychological aspects aiming at maintaining a high health-related quality of life (HRQoL) which represents a critical component in the assessment and treatment of epilepsy [2,3]. Several studies demonstrated that children with epilepsy have poorer HRQoL than healthy controls or children with other chronic health conditions [4,5,6]. Even a single seizure is associated with reduced HRQoL [7]. Several clinical and psychosocial variables influenced HRQoL in children with epilepsy including psychiatric comorbidities [8,9], psychosocial difficulties [10], cognitive functioning [11], executive dysfunction [12], and epilepsy-related variables like seizure frequency [7,13], duration of epilepsy [14], age at seizure onset [8,13], medication side effects [15,16], and the number of anti seizures medications (ASMs) [15,17].

In self-limited epilepsy with centrotemporal spikes (SeLECTS), considering the excellent prognosis for seizure remission, it is generally supposed that HRQoL is not significantly affected. However, while epilepsy is active, psychological, behavioral, and neuropsychological deficits may emerge, as variables that could negatively influence HRQoL. Regarding neurocognitive difficulties in patients with SeLECTS, the most prevalent difficulties are related to executive function (EFs) [12]. The term “EFs” refers to a set of higher-order cognitive processes mediated by neuroanatomical circuits of the prefrontal cortex (PFC) and its connections [18]. These abilities influence critical developmental areas such as academic functioning [19], emotional regulation [20], and quality of life [21]. This suggests that patients with SeLECTS might be at risk for poor HRQoL not only for their neurocognitive, emotional, and medical factors and the risk might increase if they present a poor EFs functioning. Since some demographic and epilepsy-related variables are not editable, it is remarkable to identify modifiable factors such as EFs through therapeutic interventions or accommodations to increase HRQoL.

The current study aims to:

- (1) investigate the correlation between EFs and HRQoL in a cohort of children with SeLECTS.
- (2) identify possible factors included epilepsy-related variables related to HRQoL.

2. Material and methods

The study is an observational multicenter study involving 11 Italian Referral Epilepsy Centers recognized for the diagnosis and treatment of epilepsy by the Italian Chapter of the ILAE.

This study was approved by the Ethical Committee (20210032863) of IRCCS Mondino Foundation, Pavia. Written informed consent was obtained from caregivers. Recruitment period lasted for 12 months (from January 2022 to January 2023).

2.1. Participants

Recruited patients were diagnosed with SeLECTS according to the criteria set by the ILAE Classification of the Epilepsies [22].

The inclusion criteria were a) diagnosis of SeLECTS made by a trained epileptologist in accordance with the ILAE criteria; (b) age between 3 and 14 years; (c) a stable treatment with ASMs (when applicable) from at least 4 weeks.

Exclusion criteria were (a) history of significant neurological or mental diseases (e.g., brain malformations, stroke, head injury, schizophrenia, etc.) (b) lack of parental fluency in Italian due to the written language requirements of the standardized questionnaires, (c) intellectual disability (ID) and/or borderline intellectual functioning (BIF) diagnosis (FSIQ < 85).

2.2. Measures

2.2.1. Health-related quality of life

The Pediatric Quality of Life Inventory 4.0 Generic Core Scales (PedsQL) is designed for patients aged 2–25 [23] (Varni 2009;5.), according to patients' ages, different versions were administered. The PedsQL, Italian validated form, is a questionnaire consisting of 23 items, categorized into four subscales: physical, emotional, social, and school functioning as well as three summary scores: total score, physical health summary score, and psychosocial health summary score. The physical health summary score is the same as the physical functioning subscale; to create the psychosocial health summary score the mean is computed as the sum of the items divided by the number of items answered in the emotional, social, and school functioning subscales. Scaled scores are standardized and range from 0 to 100, with higher scores indicating better quality of life. Cutoffs for this study were chosen to be based on the study of Varni [24] who put a meaningful cut-off point score for an at-risk status for impaired HRQoL for parent proxy-report score of 65.4, at one standard deviation below the average PedsQL score for healthy children. In our study the Cronbach's Alpha of subscales are: physical $\alpha = 0.811$, emotional $\alpha = 0.818$, social $\alpha = 0.807$, and school functioning $\alpha = 0.769$.

2.2.2. Parent-reported executive function

The Behavior Rating Inventory of Executive Function (BRIEF) [25,26] are standardized rating scale assessing children's EFs in everyday environments completed by parents of preschool children aged 2–5 years [25] and school children aged 5–18 years [26]. The BRIEF-P Parents' Report Form consists of 63 items comprising five clinical scales: inhibit, shift, emotional control, working memory, and plan/organize. These scales form three indexes: inhibitory self-control (ISCI; inhibit+emotional Control), flexibility (FI; shift+emotional Control), and emergent metacognition (EMI; Working Memory + Plan/Organize), and one composite score – the GEC (ISCI+FI+EMI). The BRIEF-2 Parents' Report Form consists of 63 items, with nine clinical scales: inhibit, shift, emotional control, self-monitor, initiate, working memory, plan/organize, task-monitor, and organization of materials grouped under three composite indexes: behavior regulation index (BRI; inhibit + self-monitor), emotional regulation index (ERI; shift + emotional Control), and cognitive regulation index (CRI; initiate + working memory + plan/organize + task-monitor + organization of Materials) – and one comprehensive composite summary score, the Global Executive Composite (GEC; BRI + ERI + CRI). For all BRIEF-P and BRIEF-2 clinical scales and indexes, T scores from 60 to 65 are considered mildly elevated, T scores from 65 to 69 are considered potentially clinically elevated, and T scores at or above 70 are considered clinically elevated.

2.2.3. Cognitive functioning

Cognition was measured with a Full-Scale Intelligent Quotient (FSIQ) score obtained from the Wechsler Intelligence Scale for Children, Fourth Edition (WISC-IV) [27]. This measure is an individually administered and nationally standardized measure of cognition.

2.2.4. Demographic and epilepsy-related variables

Demographic variables (e.g., gender and age) and epilepsy-related variables (e.g., age of onset, age at last seizure, epilepsy duration, seizure frequency, number of concomitant ASMs at time of assessment, treatment duration) were collected.

2.3. Statistical approach

Descriptive statistics for demographic variables and epilepsy-related variables were calculated. The correlation between BRIEF clinical scales and indexes, cognitive functioning FSIQ, and the PedsQL summary scores and subscales was tested with Pearson’s correlations. As regards the association between epilepsy-related variables and the PedsQL, Pearson’s correlations were used for the age of epilepsy onset, age at last seizures, treatment duration, and number of ASMs. For dichotomous variables (seizure frequency and ASMs) independent sample t-tests were used. After preliminary Pearson’s bivariate correlations, hierarchical linear regression models were performed to analyze the effects of epilepsy-related variables and EFs on HRQoL. Only variables that resulted significantly associated with the HRQoL in the preliminary bivariate analyses were tested as predictors in the linear regression models (stepwise method). Statistical analyses were conducted according to $p < 0.05$, using IBM SPSS 25.0.

3. Results

3.1. Sample characteristics

Our sample cohort was composed of 129 children (41 boys, mean age 119,1 months) with a diagnosis of SeLECTS. Demographic and epilepsy-related characteristics are presented in Table 1.

3.2. Neuropsychological measures

Our sample performed in the average range across all the subscales and summary scores of the PedsQL questionnaire demonstrating a good HRQoL (See Table 2). The comparison tested between the subscales scores (ANOVA $F = 27.82$; $df = 4$; $p < .001$) showed that parents reported significantly higher scores for physical health and social functioning.

Our sample performed in the normal range of the BRIEF questionnaire, showing no difficulties related to ecological EFs skills (see Table 3). The clinical scale with the lowest score, within the normal range, was Organization of Materials (ANOVA $F = 4.51$; $df = 12$; $p < .001$).

Regarding cognitive functioning, the mean FSIQ for the sample was in the average range (FSIQ = 104.45, SD = 12.62) (see Table 1). Concerning the relationship between intellectual functioning and HRQoL,

Table 1
Demographic and epilepsy related characteristics of the children with SeLECTS (n = 129).

Patients with SeLECTS n = 129	n (%) / M (SD)
Gender	male = 41 (31.8 %) female = 88 (68.2 %)
Age (months)	119.10 (28.89)
Age at seizures onset (months)	82.07 (27.93)
Age at last seizure (months)	99.76 (29.07)
Epilepsy duration (months)	36.93 (30.07)
Seizure Frequency	1–3 per year = 60.5 % (n 78) 4–9 per year = 19.4 % (n 25) ≥ 10 per year = 3.1 % (n 4)
ASMs	67 (51.9 %)
Number of ASMs	0.57 (0.75)
ASMs therapy duration (months)	28.55 (24.13)
FSIQ	104.5 (12.62) (median = 103; min = 85; max = 138)

Abbreviations: SeLECTS, Self-Limited Epilepsy with Centro-Temporal Spikes; n, number; M, mean; SD, standard deviation, ASMs, Anti-Seizure Medication; FSIQ, Full scale Intelligence Quotient; min, minimum; max, maximum.
Note. Age at last seizure was measured as the age of the child when he had the last. Epilepsy duration is calculated from the first referred seizure. ASMs therapy duration refers to the period in which the patient took the ASM.

Table 2
Scales Descriptives for PedsQL 4.0 Generic Core Scales: Parent Proxy-Report.

	n	M	SD	Min	Max
Parent proxy-report					
Total Score	126	83.21	12.59	44.56	100
Physical Health	126	87.27	14.37	25.00	100
Psychosocial Health	126	81.01	13.94	35.00	100
Emotional Functioning	126	75.78	18.43	25.00	100
Social Functioning	126	86.83	14.96	45.00	100
School Functioning	126	80.48	17.40	25.00	100

Abbreviations: n, number; M, mean; SD, standard deviation; Min, minimum; Max, maximum.

we didn’t observe a significant correlation between the two domains (see Table 4).

3.3. Correlation analyses

3.3.1. Association between EFs and HRQoL

We examined the association between the scores on the BRIEF and the level of HRQoL (see Table 4). We observed a significant association between almost all the clinical scales and the indexes with the subscales of BRIEF and the summary scores of PedsQL, except for emotional functioning and school functioning where we observed a lack of association and less consistent trend with the subscales of BRIEF. Overall, the results indicated that lower functioning in EFs was associated with lower overall HRQoL scores.

3.3.2. Relationship with epilepsy-related variables with HRQoL

We explored the relationship between epilepsy characteristics and scores on the PedsQL. (See Table 5).

We observed negative correlation between: -the duration of ASMs therapy with parent school functioning; -number of ASMs assumed with total score and school functioning subscale; -and age at last seizure with total score, psychosocial health subscale, and school functioning subscale. In addition, parents of children with a low frequency of seizures reported a better level of HRQoL in almost all the domains, aside from physical health and social functioning subscales (see Table 6). Moreover, parents of children who are not in therapy with ASMs show a better school function compared to those who take ASMs ($t = 2.18$; $d.f. = 122$; $p = 0.031$). Age at epilepsy onset and duration of epilepsy did not correlate with any of the PedsQL scales.

We observed that a longer duration of therapy, the use of ASMs, and a higher seizure frequency were found to be associated with a lower HRQoL. The school functioning on the PedsQL seemed to correlate the most with epilepsy variables.

3.3.3. Determining the HRQoL predictors: Multiple regression analyses

Based on preliminary correlations, hierarchical linear regression models were performed to analyze the effects of epilepsy-related variables and EFs on HRQoL (see Table 7). We included in the regression models BRIEF clinical scales and epilepsy-related variables (e.g., seizure frequency, age at last seizure and number of ASMs) to predict HRQoL. Regression models showed a significant negative association of BRIEF subscales plan/organize, shift, inhibition and seizure frequency with the total HRQoL score. We identified that difficulties in planning/organizing and high seizure frequency represent risk factors for the physical health score. Moreover, lower functioning in task monitor, emotional regulation, and later age at last seizure were found to have a negatively impact on psychosocial health scores. In addition, we observed that a lower functioning in the EFs domain and, specifically, shift, inhibit and working memory, represent the variables that contributed to a lower emotional functioning. Furthermore, difficulties in plan/organization, emotional regulation, and later age at the last seizure predicted worse social functioning. Lastly, task monitor, working memory, and seizure frequency showed an association with school functioning.

Table 3
Individual T-scaled scores on BRIEF-P and BRIEF-2.

	n	M	SD	Min	Max	% potentially elevated	% clinically elevated
Inhibition	126	50.9	11.98	35	91	3.17	9.5
Shift	126	52.83	13.41	37	96	11.11	9.5
Emotional control	126	51.29	13.00	36	93	6.34	7.9
Self-monitor	118	48.54	11.50	33	76	5.08	5.1
Initiate	118	50.25	10.87	35	82	7.62	4.2
Working memory	126	50.94	12.16	36	90	3.96	7.9
Plan/ Organize	126	49.72	11.53	32	94	6.34	2.4
Task-monitor	117	50.2	11.85	33	90	6.83	5.1
Organization of Materials	118	47.47	9.92	36	83	2.54	2.5
BRI	118	50.08	12.40	33	99	3.38	6.8
ERI	118	51.94	13.82	34	94	6.77	9.3
CRI	118	50.01	12.44	32	86	6.77	5.9
ISCI	8	55.75	16.35	39	88	0	12.5
FI	8	52.25	11.54	42	74	0	12.5
EMI	8	50.87	11.49	39	72	0	12.5
GEC	125	51.4	13.98	32	103	5.6	8

Abbreviations: n, number; M, mean; SD, standard deviation; Min, minimum; Max, maximum; BRI, Behavioural Regulation Index; ERI, Emotional Regulation Index; CRI, Cognitive Regulation Index; ISCI, Inhibitory self-control Index; FI, Flexibility Index; EMI, Emergent Metacognition Index; GEC, Global Executive Composite.

These relationships suggest that executive dysfunction and epilepsy-related variables could affect HRQoL. Our data suggest a major association between EFs abilities and HRQoL rather than epilepsy related variables.

4. Discussion

4.1. HRQoL

Epilepsy is a multifaceted neurological disorder with different effects on a patient’s and family’s psychosocial well-being. The neurobiological, cognitive, psychological, and social consequences of this condition are so substantial for both patients and their families that they can affect the whole lifestyle. Therefore, understanding the complex impact of epilepsy on a person’s HRQoL is widely recognized as an important component of clinical care [28].

This study aimed to examine the HRQoL in patients with SeLECTS as perceived by their parents, to investigate the potential relationship between of EFs and epilepsy related variables and HRQoL, and trying to define possible predictive variables on HRQoL in a group of patients with SeLECTS.

We reported an overall good HRQoL in children with SeLECTS as it was rated by their parents, demonstrating that children with SeLECTS had typical levels of HRQoL. This result could be linked to the self-limited nature of SeLECTS with an excellent prognosis for seizure remission, brief and infrequent seizures, good response to ASMs (when prescribed), and neurological typical functioning [29]. These results are consistent with the findings reported in the study by Bansal and colleagues [30], where the authors showed typical functioning on all the domains of the PedsQL in 115 children with self-limited epilepsy. This is in contrast to the study conducted by Connolly and colleagues [3] where a compromised level of HRQoL was found in a sample of 22 patients with SeLECTS. This might be due to the small sample size, the highest percentage of children in treatment with ASMs and the use of different HRQoL assessment measures in that study.

Regarding the PedsQL questionnaire, the domains with the higher scores are the parent physical health and social functioning subscales, we speculate that these results might be attributed to different factors such as the absence of physical restrictions and the good seizures control related to SeLECTS. Several studies demonstrated that HRQoL depends on numerous factors including: seizure frequency and control [7,13], duration of epilepsy [14], age at seizure onset [8,13], medication side effects [15,16], and the number of ASMs [15,17]. Liu and colleagues [31] demonstrated that HRQoL was influenced by age of the child with epilepsy, duration of epilepsy, seizures frequency in a sample of 223

patients with epilepsy. Moreover, in the study of Miller et al. [32] children with epilepsy showed a lower HRQoL than a control group of healthy children, for children with epilepsy, comorbid neurological impairments and number of ASMs were associated with diminished HRQoL. Regarding seizure control, Hamid et al. [33] reported that even partial seizure control improved HRQoL.

4.2. Ecological EFs

Our sample scored within the normal limits at the BRIEF-2, but the SD on the scales were more variable than typical suggesting that our sample had a higher percentage of at-risk and clinically significant ratings. These data are in contrast with other studies demonstrating several deficits on multiple scales of the BRIEF, with working memory and planning/organization being particularly concerning [13,18,34,35]. In addition, it is widely recognized that in children with SeLECTS EFs may be impaired [12]. The inconsistencies between the data might be attributed to the different kinds of assessment (direct versus indirect), the agreement between objective tests and parent-reported evaluation of EFs remains elusive.

The scale of BRIEF showing the lowest was Organization of Materials, a measure of the orderliness of personal space such as play areas, workspaces, and storage areas [26], suggesting that the abilities measured by this scale were not commonly seen as a domain of significant difficulty for the children in our sample.

4.3. The effect of EFs on HRQoL

Statistically significant associations were identified between parental-reported EFs and parent-proxy-reported HRQoL. This study provides evidence that a lower EFs control was associated with a lower HRQoL in children with SeLECTS. Our findings are congruent with other studies which found that a low performance on ecological EFs was associated with a poor HRQoL [21,34,36]. Examination of the correlations suggests that at the composite level, the indices of BRI, GEC, and ERI were related to all aspects of HRQoL. At the subscale level, Inhibit, shift, Emotional Control, Self-Monitor, and Task-Monitor were also related to all dimensions of HRQoL. Emotional functioning scale did not show a link with self-monitor, working memory, plan/organize, organization of material, task monitor and CRI. The absence of a relationship between these variables could be related to the fact that these clinical scales are more related to cognitive regulation rather than emotional regulation. EFs is having a significant role on the functioning across multiple domains in patients with epilepsy. The results provide evidence that executive dysfunction represents a significant barrier to HRQoL in

Table 4
Correlations between BRIEF and PedsQL 4.0 Generic Core Scales.

PedsQL	BRIEF scores clinical scales and composite indexes													WISC
	I	S	EC	SM	In	WM	P/O	TM	OM	BRI	ERI	CRI	GEC	
TS	-0.351**	-0.423**	-0.325**	-0.358**	-0.307**	-0.310**	-0.394**	-0.446**	-0.281**	-0.335**	-0.379**	-0.354**	-0.388**	0.019
PH	-0.261**	-0.333**	-0.257**	-0.279**	-0.276**	-0.255**	-0.356**	-0.290**	-0.260**	-0.268**	-0.309**	-0.284**	-0.299**	0.056
PSH	-0.339**	-0.403**	-0.307**	-0.344**	-0.274**	-0.289**	-0.351**	-0.459**	-0.245**	-0.316**	-0.353**	-0.335**	-0.372**	-0.008
EMF	-0.251**	-0.312**	-0.220**	-0.197**	-0.133	-0.033	-0.141	-0.254**	-0.11	-0.202*	-0.265**	-0.149	-0.227*	-0.120
SF	-0.199*	-0.294**	-0.327**	-0.327**	-0.298**	-0.268**	-0.327**	-0.358**	-0.156	-0.231*	-0.299**	-0.278**	-0.333**	0.016
SCF	-0.377**	-0.384**	-0.223*	-0.344**	-0.265**	-0.430**	-0.414**	-0.537**	-0.345**	-0.352**	-0.318**	-0.416**	-0.370**	0.100

Note. * $p < .05$; ** $p < .01$;
Abbreviations: SS, subscales and Summary scores; TS, Total Score; PH, Physical Health; PSH, Psychosocial Health; EMF, Emotional Functioning; SF, Social Functioning; SCF, School Functioning; I, Inhibit.; S, shift; EC, emotional control; SM, self monitor; In, Initiate; WM, Working Memory; P/O, Plan/Organize; TM, Task-Monitor; OM, Organization of Materials; BRi, Behavioral Regulation Index; ERI, Emotion Regulation Index; CRI, Cognitive Regulation Index; ISCI, Inhibitory Self-Control Index; FI, Flexibility Index; GEC, Global Executive Composite; TS, Total score PedsQL; PH, Physical health; PSH, Psychosocial health; EMF, Emotional functioning; SF, Social functioning; WISC-IV, Wechsler Intelligence Scale for Children, Fourth Edition; FSIQ, Full scale Intelligent Quotient.

Table 5
Correlations between epilepsy-related variables with PedsQL 4.0 Generic Core Scales.

PedsQL subscales and summary scores	Age first seizure	Age last seizure	Epilepsy Duration	Number ASMs	Duration of therapy
Total Score	-0.103	-.210*	-0.06	-.194*	-0.106
Physical Health	0.296	0.033	0.543	0.049	0.489
Psychosocial	-0.084	-0.161	0.019	-0.182	0.07
Health	0.395	0.103	0.847	0.065	0.649
Emotional	-0.1	-0.206*	-0.095	-0.175	-0.183
Functioning	0.312	0.036	0.34	0.076	0.228
Social	-0.119	-0.191	-0.053	-0.114	-0.032
Functioning	0.231	0.052	0.594	0.247	0.835
School	-0.046	-0.087	-0.038	-0.098	-0.05
Functioning	0.645	0.378	0.701	0.322	0.744
School	-0.08	-0.223*	-0.14	-0.211*	-0.348*
Functioning	0.418	0.023	0.155	0.032	0.019

Note. * $p < .05$; ** $p < .01$;
Abbreviations: ASMs, Anti-Seizures Medications.

Table 6
Seizure frequency and PedsQL 4.0 Generic Core Scales Parent Proxy-Report.

	Mean (SD)		t	d.f.	p-value
	Group a (1–3 seizures per year)	Group b (4–10 seizures per year)			
Total Score	84.65 (12.38)	77.16 (13.19)	2.55	98	0.012
Physical Health	88.50 (14.02)	84.50 (14.30)	1.22	98	ns
Psychosocial	82.59 (13.38)	73.24 (15.35)	2.91	98	0.004
Health					
Emotional	77.18 (17.40)	65.20 (20.48)	2.85	98	0.005
Functioning					
Social	87.40 (15.40)	83.60 (15.24)	1.07	98	ns
Functioning					
School	83.20 (15.80)	71.20 (22.51)	2.47	32.25	0.019
Functioning					

Abbreviations: M, mean; SD, standard deviations; t, t-test;; d.f., degrees of freedom.

children with SeLECTS. EFs showed a significant impact on the day-to-day experience and functioning contributing to HRQoL in patients with epilepsy [21,34,36]. Taking into consideration that certain epilepsy-related variables could not be modified, the possibility to identify skills sensitive and modifiable through neurocognitive and psychological treatment represents a promising target of transdiagnostic therapeutic intervention [37]. Several studies demonstrated the possibility to improve EFs skills [38] emphasizing the importance of practicing EFs skills in a variety of contexts and also encouraging children to actively reflect on EFs to learn the purpose and utility of EF skills [38].

4.4. The effect of an epilepsy-related variable on HRQoL

We did not find a relationship between **age at epilepsy onset** (mean age of 6,8 years) and **HRQoL** as was already found in earlier works that considered this variable [13,14,34]. In contrast, some studies reported a correlation between early age at seizure onset and a low HRQoL [21,39].

We did not observe an association between the **duration of epilepsy** with any of the PedsQL scales. The effect of the duration of the disease on the HRQoL of patients with epilepsy represents a controversial issue. On one hand, some studies observed that a longer duration of epilepsy negatively influences the HRQoL of the patients representing the weariness and the increased burden that chronic diseases have on patients [40]. On the other hand, other studies have found improvements in HRQoL in chronic epilepsy after initial diagnosis that may be attributable to family adjustment [41,42].

We postulate that the absence of a relationship between HRQoL and

Table 7

Linear regression models.

Total Score	R2	f	Sig	B	Beta st	CI-LL	CI-UL	p-value
Plan/ Organize	0.497	23.69	0.000	−0.400	−0.310	−0.687	−0.112	0.007
Shift				−0.348	−0.296	−0.585	−0.112	0.004
Seizure frequency				−4.373	−0.181	−7.94	−0.801	0.017
Inhibition				−0.260	−0.205	−0.504	−0.016	0.037
Physical health	R2	f	Sig	B	Beta st	CI-LL	CI-UL	p-value
Plan/ Organize	0.288	19.616	0.000	−0.808	−0.531	−1.074	−0.541	0.000
Seizure frequency				−5.295	−0.185	−10.298	−0.291	0.038
Psychosocial health	R2	f	Sig	B	Beta st	CI-LL	CI-UL	p-value
Task monitor	0.51	32.885	0.000	−0.616	−0.486	−0.817	−414	0.000
Emotional Control				−0.464	−0.352	−0.671	−0.256	0.000
Age last seizure				−0.102	−0.180	−0.187	−0.017	0.019
Emotional functioning	R2	f	Sig	B	Beta st	CI-LL	CI-UL	p-value
Shift	0.376	19.508	0.000	−0.876	−0.520	−1.235	−0.516	0.000
Inhibition				−0.691	−0.380	−1.061	−0.322	0.000
Working memory				0.560	0.297	0.148	0.972	0.008
Social functioning	R2	f	Sig	B	Beta st	CI-LL	CI-UL	p-value
Plan/ Organize	0.372	19.171	0.000	−0.499	−0.338	−0.794	−0.204	0.001
Emotional Control				−0.488	−0.353	−0.766	−0.211	0.001
Age last seizure				−0.106	−0.178	−0.205	−0.007	0.037
School functioning	R2	f	Sig	B	Beta st	CI-LL	CI-UL	p-value
Task monitor	0.524	34.753	0.000	−0.618	−0.388	−0.948	−0.288	0.000
Working memory				−0.693	−0.383	−1.065	−0.320	0.000
Seizure frequency				−5.805	−0.175	−10752	0.858	0.022

Abbreviations: R2, R-squared (regression coefficient); f, F-test for overall significance; Sig, Significance; B, unstandardized beta; Beta st, standardized beta; CI-LL, Confidence Interval- Lower Limit; CI-UL, Confidence Interval-Upper Limit.

the age of seizure onset and the duration of the epilepsy could be due to the self-limited nature of SeLECTS.

In addition, in our study parents of children with a **low frequency of seizures** reported a better level of HRQoL in almost all the domains, aside from social functioning subscales. These findings are consistent with other studies proving seizure frequency to be a significant inverse predictor of HRQoL across different domains [14,31]. Specifically, in the study of Modi et al. [43] the authors demonstrated a global HRQoL deficit in children with a single seizure. These results suggest that it is the seizure event that may affect the HRQoL rather than the diagnosis of epilepsy. The main characteristic of epilepsy seizure is the unpredictability causing anxiety in parents, therefore HRQoL might be affected by parental anxiety [44].

We reported an association between **age at last seizure** (mean age 8,3 years) with total score, psychosocial health, and school functioning subscales. A prolonged period characterized by seizure freedom showed a positive correlation with HRQoL, suggesting that seizure reduction contributes to an improvement of HRQoL, which is a well-supported finding within the literature [45,46]. These data can be interpreted taking into consideration that reduction and seizure freedom are commonly considered foremost achievements in the treatment of epilepsy both for clinicians and families.

Regarding **epilepsy treatment**, we reported a negative association between the number of ASMs [47] with total score and school functioning subscales. A negative association between **treatment duration** with school functioning subscales was found; moreover, there was a significant difference in favor of **children untreated**, who show a better school functioning compared to those who take ASMs. Based on the data collected in our study a longer duration of therapy showed a negative association with the HRQoL of children with SeLECTS. Our data suggest

a negative association between ASMs and the HRQoL of patients with SeLECTS. Our result could be attributed to the adverse effects of medication [48]. In fact, adverse cognitive effects of ASMs are associated with reduced HRQoL and treatment non-adherence [49,50]. Several studies reported a negative influence of ASMs' cognitive, behavioral, and HRQoL [51,52]. On the other hand, some ASMs are associated with enhanced cognitive, emotional, and HRQoL [51,52]. International League Against Epilepsy (ILAE) Task Force Report concluded that there was a notable lack of information on the cognitive and behavioral effects of ASMs in children and adolescents and further prospective randomized studies are needed to better elucidate ASMs influence on patients with epilepsy [53]. On the other hand, the higher the number of ASMs the lower the score obtained on HRQoL, this could be also attributed to the presence of different levels of disease severity: when the epilepsy is more severe clinicians are more likely to introduce ASMs, and this datum might make the difference between treated and untreated patients. Secondly, the adverse effects of ASMs might cause neurocognitive deficits affecting HRQoL due to their inherent neuromodulatory actions [52]. ASMs reduce neuronal irritability but also reduce neuronal excitability interfering with normal neuronal networks and, so, may induce cognitive deficits [54]. In general, the wide variability in the effects of medications makes it difficult to come to certain conclusions and attribute changes in cognitive, behavioral, and HRQoL to the presence or lack of treatment. Thus, it is very important to take into consideration the risk/benefit ratios during treatment decisions.

4.5. Risk factors of HRQoL

Finally, we wanted to detect the risk factors for a low HRQoL to identify patients with a high level of risk of a low QoL. We observed that

executive dysfunction was a significant predictor of reduced HRQoL, equivalent to epilepsy-related predictors such as **seizure frequency** and **age at the last seizure**. Our observations suggest that lower ability related to EFs represents a significant barrier to the HRQoL influencing several components of HRQoL. Moreover, the presence of executive dysfunction, higher seizure frequency, and recent seizures appeared to contribute significantly to the risk of poor HRQoL in children with SeLECTS. These results demonstrated the association between EFs and epilepsy-related variables in psychological development in the broadest sense, including behavioral regulation, social knowledge integration, and academic achievement.

5. Conclusion

In conclusion, our results suggest the importance of the identification of patients with SeLECTS with a high level of risk for a poor HRQoL. Moreover, we may now add executive dysfunction to the list of known risk factors for poor HRQoL in children with SeLECTS, along with factors such as seizure frequency, recent seizures, use of ASMs, longer duration of therapy. The early identification of children with SeLECTS at risk of a poor HRQoL may allow the activation of interventions: while it is not possible to modify epilepsy-related variables, the possibility to work with EFs' sensitive and modifiable skills represents a promising target of therapeutic intervention to improve HRQoL in children with epilepsy. Improving HRQoL should represent the most important outcome variable in patients with epilepsy. Therefore, studies that identify predictors of HRQoL appear to be crucial for the identification and planning of interventions aimed at ameliorating clinical conditions in epilepsy patients. The evidence that both cognitive and epilepsy-related variables influence HRQoL confirm that patients with epilepsy require holistic care.

This study has some limitations: firstly, the two primary measures (the BRIEF and PedsQL) both involve parent reports; therefore, the observations obtained from the study might be limited to the parent's perceptions of health and neuropsychological status of the patients which could differ from the patient's perceptions. Therefore, future studies of HRQoL in children with epilepsy may consider self-reported measures and the use of direct measures to assess EFs.

The second limitation concerns the fact that HRQoL was assessed using a generic questionnaire, future studies could include tailored questionnaires for patients with epilepsy, validated in different local languages.

Furthermore, while epilepsy-related variables have been considered potential contributors to HRQoL, biopsychosocial factors have not been included and are likely to play a significant role in HRQoL.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Italian Ministry of Health RC 2022 and by ERN-Epicare. We are truly grateful to our collaborators of the **EpilEpsyIt study group**: Rosaria Calabrese, Marco Fasce, Gaia Garghetti, Lucio Giordano, Antonella Morea, Miriam Olivieri, Grazia Papalia, Maria Valvassori, Aglaia Vignoli, Maurizio Viri, Arianna Zaliani.

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